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The Palaeomagnetism of the Great Dyke of Southern Rhodesia

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Summary

Oriented cores have been secured from fourteen sites in the Great Rhodesian Dyke, by means of a portable sampling drill. The natural remanent magnetizations showed high dispersion at all sites except one. After demagnetization in alternating magnetic fields, nine sites gave well grouped directions of primary magnetization. These sites include five rock types distributed among three Complexes of the Great Dyke and two satellite dykes, over 200 miles of the length of the Dyke and through several thousand feet in depth as the rocks were originally intruded. The nine site mean directions of primary magnetization are closely grouped and are believed to represent directions of thermo-remanent magnetization at the date of intrusion of the Great Dyke. It is suggested that the dates of magnetization at the sites must cover a sufficient time interval to give a mean pole position close to the axial geocentric dipole freed from secular variation. On the assumption of a geocentric dipole field, the position of the mean South magnetic pole is $21\frac{1}{2}$ °N, $61\frac{1}{2}$ °E, with radius of 95 per cent confidence 9°. This pole position is close to positions of North magnetic poles given by studies of the palaeomagnetism of the Pilansberg Dykes and Bushveld gabbro.

1. Geology of the Great Dyke

The great Dyke of Southern Rhodesia has been described in detail by Worst (1958, 1960). The latter publication includes a detailed geological map in colour on a scale $1:10^5$. According to Worst the Great Dyke consists of the remains of four lopoliths which were formed along a straight line and may have intersected one another. The four complexes have been partly preserved in a graben and the

outline of a dyke has resulted on the surface now exposed by erosion. Figure 1 shows the Great Dyke and its division into four lopoliths, named by Worst the Musengezi, Hartley, Selukwe and Wedza complexes. The overall length of the Dyke is 332 miles, and its width ranges from 1·5 to 6·5 miles, the average width being 3·6 miles.

The Great Dyke is intrusive into ancient granite over 97 per cent of its length, and the marginal contacts dip rather steeply toward the axis of the Dyke. The four complexes exhibit a marked layering of the component rock types. These layers show, on each side of the dyke, moderate dips towards the centre of the dyke, in a general synclinal structure. These dips are generally smaller than the dips of the granite contacts. The rock types range from ultramafic at depth to mafic in the uppermost layers. Boreholes indicate dunite at depths of a few thousand feet overlain by a succession of differentiated layers of harzburgite, picrite, olivine-pyroxenite and pyroxenite, interleaved with chromite seams. The olivine-bearing rocks are largely altered to serpentine at the surface, and no appreciable outcrops of dunite are found.

Remnants of gabbroic rocks overlie the uppermost pyroxenite bands in all four complexes. Worst considers that each complex was intruded through a single feeder,

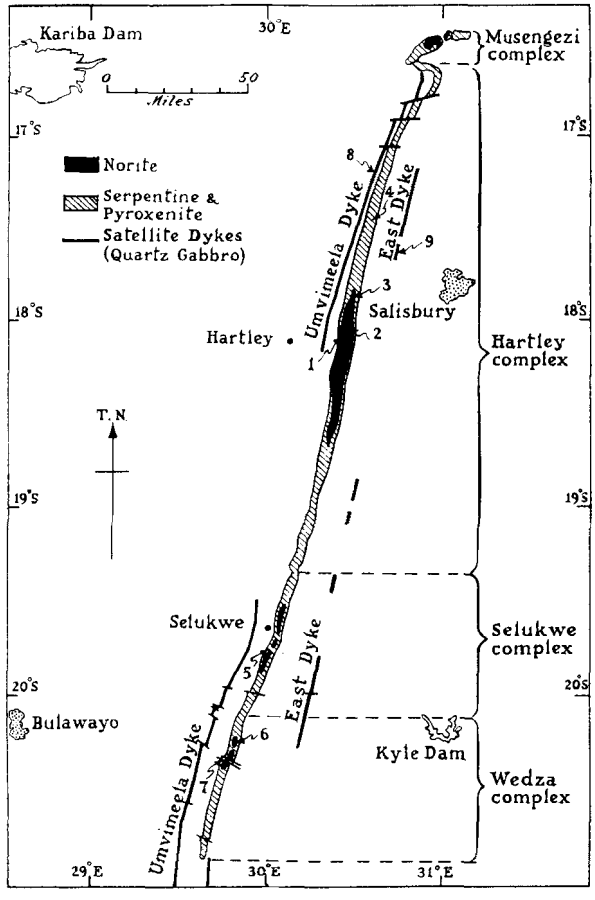


FIG. 1.—The Great Dyke of Southern Rhodesia. The positions of nine sampling sites are indicated.

whose position is approximately indicated by the outcrops of the gabbroic rocks in each complex. The gabbroic rocks are mainly norite, and are so named in Figure 1.

Worst considers that the layers in the Dyke rocks were formed horizontally as a result of successive magma injections and differentiation, and that the synclinal deformation developed concurrently with the downward displacement of the Great Dyke involved in the graben formation. Longitudinal faults flanking the graben are present in the granite, but not in the Dyke rocks. This leads Worst to suggest that the graben was formed, and the synclinal deformation of the Dyke arose, while the Dyke rocks were plastic.

Two satellite dykes of quartz gabbro are intermittently exposed on either side of the Great Dyke along much of its length (Figure 1). Worst regards these dykes as intrusions into tension faults produced during the graben formation after intrusion of the Great Dyke complexes. Whatever the detailed mechanism, there is little doubt that the satellite dykes are genetically related to the Great Dyke, and that their date of intrusion is indistinguishable, on the geological time scale, from that of the Great Dyke. It is on this assumption that these satellite dykes have been sampled in the present study. It may be remarked that quartz gabbro has been preserved in a down-faulted block of the Greak Dyke where the Mtshingwe River crosses it. This block lies between two faults shown near Site 7 in Figure 1. The quartz gabbro lies on top of norite and is regarded by Worst (1960) as the final intrusion. Worst therefore correlates the quartz gabbro satellite dykes with the end of the Great Dyke series of intrusions.

Transverse faults of later date have displaced the Great Dyke at a number of points. The most important transverse faults are shown in Figure 1. Considering the age and length of the Dyke, tectonic disturbance has been remarkably slight since its intrusion. It appears certain that no appreciable rotation of the Dyke relative to the Rhodesian shield has occurred since it was intruded.

2. Aims of the investigation

The primary aim of the study was to establish a pole position for the date of intrusion of the Great Dyke. Many geologists consider that the Great Dyke is related to the Bushveld Igneous Complex, and may have been intruded contemporaneously. This study was therefore undertaken as a sequel to the palaeomagnetic study by Gough and van Niekerk (1959) of the Bushveld gabbro. In pursuance of this primary aim, sites were sampled from all available unaltered rock types, over the maximum accessible length and depth of the original intrusion. A secondary aim was to examine the question whether magnetization occurred before or after the deformation (Gough and van Niekerk 1959). To this end, efforts were made to sample synclinally deformed layers on both flanks in order to apply a fold test. These efforts were only partially successful because of the rarity of good exposures.

3. Description of sites and sampling procedure

Fourteen sites were sampled. Nine sites, distributed over some 200 miles of the length of the Great Dyke, gave useful palaeomagnetic data and are indicated in Figure 1. Sites 1 to 4 lie in the Hartley Complex, Site 5 in the Selukwe Complex

and sites 6 and 7 in the Wedza Complex. The Musengezi Complex lies in inaccessible country and was not sampled. Sites 8 and 9 are in the Umvimeela and East satellite dykes respectively, on either side of the Hartley Complex.

At each of these nine sites, cores one inch in diameter and not more than six inches in length were drilled by means of a Graham-Keiller sampling drill (Graham and Keiller 1960). Each standing core was oriented and inscribed with a diamond scratch parallel to the axis, by means of an orienting instrument to be described elsewhere. This instrument together with a theodolite enables cores to be oriented with reference to the Sun's azimuth (or any other known azimuth) within one degree, which is the uncertainty in the spinner magnetometer measurement. Some cores at Site 6 were oriented with the aid of a Brunton compass which was checked against the Sun at the same place. All other cores were oriented with reference to the Sun. At each site cores were drilled over the full exposure available, and were spaced at least six feet apart and usually more, so that each core was an independent sample and any lightning effects would not be the same for any two cores. In general, two or three cylindrical specimens one inch long were cut from each core in the laboratory. In the statistical analysis, the mean direction of magnetization of two specimens from a core is regarded as a single observation.

In the Hartley Complex Sites 1, 2 and 3 are in the largest norite exposure of the Dyke. They are located in a river bed, a road quarry and a rail cutting respectively. Site 4 is in olivine-pyroxenite exposed in a road cutting. On Worst's reconstruction of the original intrusion, Site 4 was about 4000 ft lower than Sites 1 and 2, which are near the highest existing norite horizon. Site 3 is near the base of the norite and was about 1500 ft lower than Sites 1 and 2 in the original intrusion. In the Selukwe Complex Site 5 is in harzburgite exposed in a river bed near the west flank of the Dyke. In the Wedza Complex Site 6 is in norite exposed on a hilltop. In view of the high probability (Graham 1961) that lightning had affected some outcrops at this site, three cores were taken from each of five outcrops spread over a distance of about a mile. Site 7 is just south of the Mtshingwe River near the west margin of the Dyke and just south of the Mtshingwe Fault. Ten cores were taken, of which seven from one outcrop proved to be picrite and three, from a second outcrop one-quarter mile away, proved to be quartz gabbro. Using Worst's reconstruction of the original intrusion, Site 7 was originally about 2300 ft lower than Site 6.

In the Umvimeela dyke (quartz gabbro) Site 8 is in a declivity between two hills formed by the dyke. In the East dyke (quartz gabbro) Site 9 is in a pair of small quarries.

In the Hartley Complex two sites in steeply dipping pyroxenite, one on each side of the synclinal axis, gave no useful palaeomagnetic data either before or after alternating field demagnetization. At one of these sites a thin section shows extensive alteration to serpentine with considerable secondary magnetite. At the other site a thin section shows traces of secondary magnetite. A third site in the Hartley Complex was located in norite at the Umsweswe River. The NRM directions which were well grouped and changed very little on a.c. demagnetization, differed widely from the other Great Dyke sites and from the present geomagnetic field. The anomalous magnetization is attributed to reheating by a large dolerite sill of which remnants remain in the area. This sill forms part of the subject matter of a separate investigation.

In the Selukwe Complex one site was located in harzburgite on the opposite side of the Dyke from Site 5. This harzburgite proved to be serpentinized with secondary magnetite.

In the Wedza Complex 25 cores were drilled in a good exposure at the Mtshingwe River bridge. This exposure is mapped by Worst as quartz gabbro. Fourteen coarse-grained cores were found, in thin section, to be altered quartz gabbro, and were anomalously magnetized. The other eleven cores proved to be fine-grained olivine gabbro, and may have been taken from a large xenolith. These cores were magnetically unstable.

4. The natural remanent magnetizations

The directions and intensities of remanent magnetization were measured by means of a spinner type magnetometer to be described elsewhere. Directions were measured with an uncertainty of one degree and intensities within 5 per cent. The parameters of the NRM at each site are presented in Table 1. At all sites the directions of NRM are very scattered, as will be seen from the low values of Fisher's k (Fisher 1953). The highest value of k is 7·8 at Site 7. At Site 1 and Site 6 the resultant R of the N vectors is less than the value R_0 tabulated by Watson (1956), which is exceeded in 5 per cent of cases by the resultant of N randomly directed vectors. At these two sites the distribution may, therefore, be considered random on Watson's criterion. If there is a consistent primary magnetization, it is clear that the rocks of the Great Dyke have been disturbed, in general, by secondary magnetizations.

5. Demagnetization by alternating magnetic fields

It has been shown by various workers (As & Zijderveld 1958, Creer 1959, Irving, Stott & Ward 1961) that secondary components of magnetization may often be removed, without completely destroying the primary magnetization, by a technique which is sometimes referred to as "a.c. washing". Essentially the procedure is to subject a rock specimen to an alternating magnetic field which is smoothly reduced to zero, in the absence of any steady field. The specimen is rotated nearly randomly about two axes throughout the process, and the alternating field is reduced over a time interval during which the specimen is rotated many times about each axis. The periods of rotation about the two axes are, in turn, compared with the period of the alternating magnetic field.

In the demagnetizing apparatus used in this work, rock specimens up to 3·5 cm in diameter and 3·5 cm long are placed in a cradle similar to that described by Creer (1959), which rotates about the vertical axis at 140 revolutions per minute and about a horizontal axis at 16/15 of this. The spinning specimen is at the centres of two Helmholtz pairs of coils of radii 30 cm and 34 cm, which annul the horizontal and vertical components respectively of the geomagnetic field. The alternating field is provided by a solenoid similar to that described by Irving, Stott & Ward (1961), which produces a peak field of 162 $\pm$ 3 Oe rms per ampere as measured by means of a search coil. The maximum peak field used is 800 Oe. The coil is tuned to the 50 c/s mains frequency through a bank of capacitors. A series resonant circuit tuned to 100 c/s is connected in parallel. Smooth reduction of the solenoid current is arranged by means of an electrolytic potential divider. A glass tube 6 cm in diameter and 1·3 m long is mounted vertically and has cylindrical electrodes at the ends, connected to the output of a Variac. A third electrode of smaller diameter is suspended on a P.V.C. covered wire and is raised

smoothly up the tube, which is filled with a solution of zinc chloride and hydrochloric acid. The electrolyte concentration is chosen so that the impedance of the liquid column is roughly equal to that of the tuned solenoid circuit, which is connected between the moving electrode and the upper fixed electrode. The P.V.C. covered wire is passed over pulleys and once round a conical drive shaft to a counterweight. A thread on the conical drive shaft prevents slipping. The drive shaft is driven by an electric motor through a worm gear and three-speed gearbox, which enables the moving electrode to be raised through the length of the tube in either two, five or ten minutes. The wire is started on the narrow end of the drive shaft, whose conical shape results in a nearly constant time rate of fall of the solenoid current. This form of alternating field demagnetizer has advantages of compactness and speed of operation.

With the geomagnetic field annulled and the specimen spinning, the solenoid current is raised by means of the variac to give the desired maximum peak field. The demagnetizing field is then reduced to zero by the liquid potential divider. For

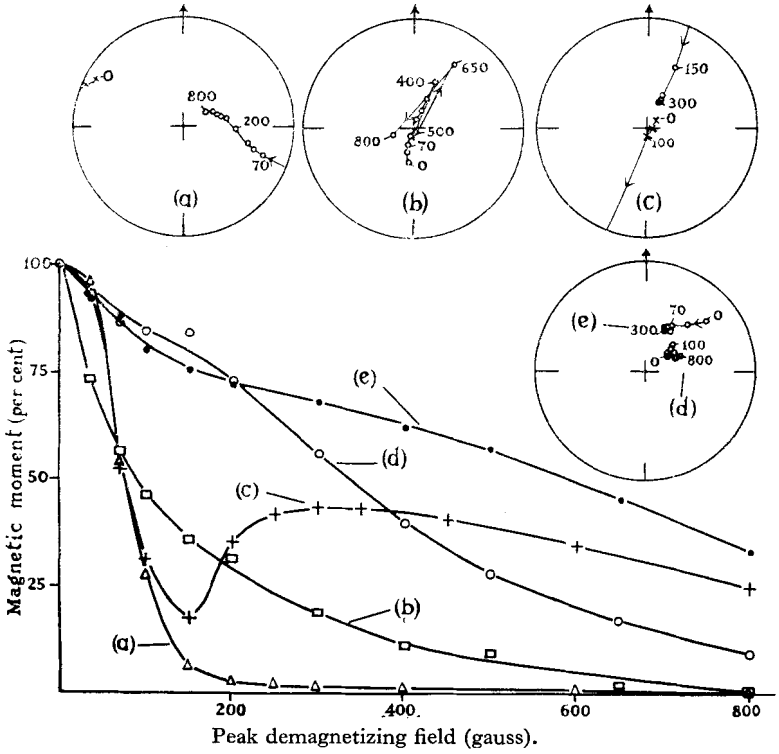


FIG. 2.—Demagnetization of specimens from four sites.

● Site 1 (Norite) — $M_n = 9{\cdot}47 \times 10^{-4}$
○ Site 7 (Picrite) — $M_n = 5{\cdot}52 \times 10^{-4}$
\+ Site 8 (Quartz gabbro) — $M_n = 7{\cdot}40 \times 10^{-3}$
□ Site 5 (Harzburgite) — $M_n = 2{\cdot}73 \times 10^{-4}$
△ Site 7 (Quartz gabbro) — $M_n = 1{\cdot}86 \times 10^{-3}$

M_n is intensity of the natural remanent magnetization in e.m.u. cm^{-3}. Directions of magnetization shown are in lower hemisphere stereographic projections, together with the peak alternating field corresponding to some points.

○—South seeking pole
×—North seeking pole
} in lower hemisphere.

peak fields up to 200 Oe, the reduction time is usually two minutes, from 200 to 500 Oe five minutes, and from 500 to 800 Oe ten minutes.

Secondary isothermal magnetizations arise in two principal ways. Lightning currents may give rise to "hard" I.R.M. of relatively high coercivity (Cox 1961, Graham 1961). "Viscous" I.R.M. of low coercivity is believed to arise from the presence of either single or multi-domain grains of relaxation time less than the age of the rock (Irving, Stott & Ward 1961). An empirical approach is therefore necessary in choosing the appropriate demagnetizing field for specimens from a given site. The usual procedure has been to demagnetize a specimen from maximum peak fields of 35, 70, 100, 150, 200, 300, 400, 500, 650 and 800 Oe in turn, the remanent magnetization being measured after each demagnetization. Figure 2 shows changes of direction and intensity of magnetization with increasing alternating field, for five representative specimens. In general, the direction of magnetization moves in a plane containing the directions of the primary component and the N.R.M., i.e. along a great circle of the sphere of projection. If, after initial changes, the direction of magnetization remains nearly constant with further increase of the maximum demagnetizing field, it is provisionally assumed that a stable primary component of magnetization is present. In Figure 2, specimens (c), (d) and (e) typify this behaviour. After reaching their end-points, these specimens show negligible further changes of direction of magnetization up to 800 Oe peak field. Specimen (c) had a soft component of magnetization almost opposed in direction to the primary magnetization, and more than twice as large. During demagnetization the direction of magnetization moved through almost two right angles, and the demagnetizing curve has a minimum where the hard and soft components were nearly equal and opposite.

When a stable end-point is reached, as has been the case at five of the nine sites used in the present work, a demagnetizing field is chosen slightly larger than the minimum field for which the trial specimen reached its end-point, and two or three specimens are washed in this field. If the directions of magnetization are then closely grouped, the existence of stable primary magnetization is confirmed, and specimens from all cores taken at the site are washed in this maximum demagnetizing field.

After treatment in high alternating fields some specimens show large irregular changes in direction of magnetization (Figure 2(b)). This effect has been discussed by Irving, Stott & Ward (1961), who suggest that it arises as a result of the randomization by the demagnetization procedure, of directions of saturation magnetization in domains. On this view the irregular changes in direction of magnetization set in when the intensity of magnetization approaches the minimum value possible in the resultant of the statistical population of domain magnetizations present in the specimen. The authors have found, as did Irving, Stott & Ward, that the irregular changes of direction are correlated with fluctuations in intensity. This supports the hypothesis quoted. In some cases the irregular changes of direction begin without any considerable pause near an end-point. In such cases the choice of a demagnetizing field is difficult. It may be necessary to wash several specimens in various fields just below the highest field for which the trial specimen remained on its great circle arc. A washing field is then selected which gives best grouping.

In Figure 2, specimen (a) illustrates the behaviour of a specimen having I.R.M. due to a lightning discharge. Neighbouring cores from the same outcrop reached end-points at less than 300 Oe, whereas specimen (a) fell in the group when demagnetized in 800 Oe.

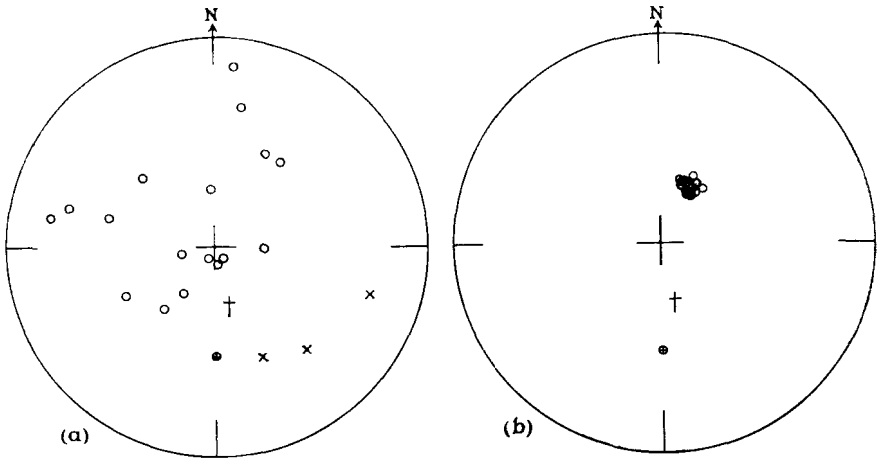


FIG. 3.—Lower hemisphere stereographic projection of directions of magnetization at Site 9 (quartz-gabbro).
(a) NRM and (b) after demagnetization in an alternating field of 400 gauss peak.
○—South seeking pole } in lower hemisphere.
×—North seeking pole }
⊕—Axial dipole field (south-seeking).
†—Present field (south seeking).

Figure 3 shows stereographic plots of the directions of N.R.M. at Site 9, and the corresponding primary magnetizations after demagnetization in peak field 400 Oe. The Fisher k parameter increases from 2·5 to 317 in this case. As will be seen in Table 1, there is a marked increase of k on demagnetization at each site used.

6. The primary magnetizations

Figure 4 shows stereographic projections of the directions of primary magnetization at Sites 1 to 8 after demagnetization in the maximum fields indicated. The parameters for the mean primary magnetizations are given in Table 1.

At Site 1 all 7 cores were used in computing the parameters of the mean direction. One was an experimental core 2 ft long drilled with the large sampling drill which will be described elsewhere. Thirteen specimens from this core have been combined to give a single core mean direction, which was given unit weight. The primary magnetization reached a stable direction for peak demagnetizing fields 200 gauss $< H_{AC} \leqslant$ 800 Oe (cf. Figure 2(e)).

At Site 2, 14 cores were drilled from small isolated rock masses in a gravel quarry. At the time of sampling, it was noted that 7 of these rock masses were doubtful in the sense that they might have been moved in the quarrying operations. Three of the doubtful cores were found to have anomalous directions of primary magnetization after alternating-field demagnetization. The primary magnetizations of the other 11 cores are represented in Figure 4(b) and in Table 1. Under demagnetization, the specimens from this site did not reach a stable direction of magnetization, and care was required in selecting the washing field of 150 Oe peak.

At Site 3 two of the 9 cores drilled gave anomalous directions of primary magnetization. The cores at this site were taken from small exposures in a shallow railway cutting, and it is probable that the anomalous cores were drilled from rolled boulders. The primary magnetization was stable for 100 Oe $< H_{AC} \leqslant$ 300 Oe.

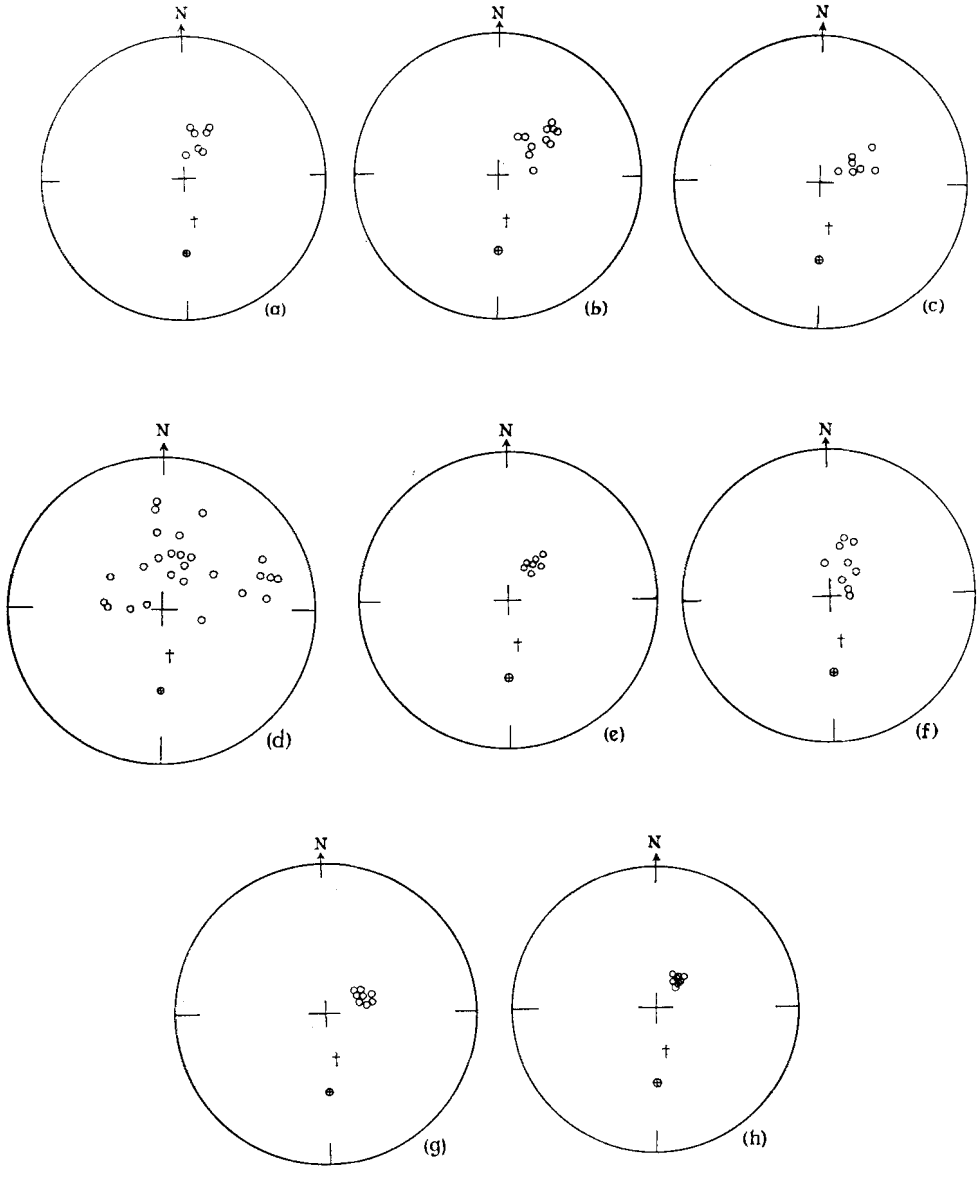


FIG. 4.—Directions of primary magnetizations at Sites 1 to 8 after demagnetization in the peak alternating fields indicated below. Lower hemisphere stereographic projection.

○—South seeking pole in lower hemisphere

⊕—Axial dipole field (south-seeking)

†—Present field (south-seeking)

(a) Site 1 (Norite)—300 Oe
(b) Site 2 (Norite)—150 Oe
(c) Site 3 (Norite)—150 Oe
(d) Site 4 (Olivine–pyroxenite)—150 Oe
(e) Site 5 (Harzburgite)—300 Oe
(f) Site 6 (Norite)—600 Oe
(g) Site 7 (Picrite and quartz gabbro)—300 Oe
(h) Site 8 (Quartz-gabbro)—300 Oe.

Table 1

Parameters of natural remanent magnetization and of primary magnetization at nine sites in the Great Dyke.

Sampling sites		Lat. S	Long. E	N	N'	E	N.R.M. D	N.R.M. I	N.R.M. R	N.R.M. k	H_{AC} Oe	Primary magnetization D	Primary magnetization I	Primary magnetization R	Primary magnetization k	α_{95}	Ancient South pole Lat. N	Ancient South pole Long. E
Hartley complex	1. Norite, Serüi River	18°04′	30°26′	7	24	0	251	− 9	2·31*	1·3	300	200½	−57½	6·893	56	8	30½	49
	2. Norite, Road Quarry, Selous	18°02′	30°30′	11	21	3	255	−33	5·84	2·4	150	228½	−48½	10·74	38	7½	23	76
	3. Norite, Rail Cutting, Kutama	17°54′	30°29′	7	13	2	305	−37½	5·72	4·7	150	242	−59	6·883	51	8½	8½	74
	4. Olivine–Pyroxenite, Road cutting, Sinoia Road	17°32′	30°35′	26	47	0	183	−64½	11·31	1·7	150	205½	−53	20·76	4·8	12½	33½	56
Selukwe complex	5. Harzburgite, Moonies Creek	19°46′	30°00′	7	14	2	256	−68	5·70	4·6	300	215	−57	6·961	154	5	24	60
Wedza complex	6. Norite, Wedza Farm	20°16′	29°49′	9	18	6	299½	−41	3·33*	1·4	600	206½	−65	8·748	32	9½	19	48
	7. Picrite and Quartz Gabbro, Sitedzi River	20°20′	29°44′	8	8	2	265½	−56	7·10	7·8	300	245	−60½	7·962	185	4	4	73
Satellite dykes	8. Quartz Gabbro, Umvimeela Dyke	17°17′	30°36′	10	10	0	221	−53	7·54	3·7	300	217	−62	9·981	474	2	20	58½
	9. Quartz Gabbro, East Dyke	17°38′	30°45′	20	20	0	134	−55	12·36	2·5	400	206½	−57½	19·94	317	2	29	54½

N = No. of cores (i.e. samples) used.
N' = No. of specimens.
E = No. of cores excluded.
H_{AC} = Peak demagnetizing field in Oe.
* = Resultant not significant at 95 per cent level (Watson 1956).

D = Declination (east of true north)
I = Inclination (positive down)
} of North seeking pole.
R = Resultant of N unit vectors.
k = $(N-1)/(N-R.)$
α_{95} = Semi-angle of cone of 95 per cent confidence.

At Site 4 some 26 cores were drilled from olivine–pyroxenite exposed, six months after blasting, in a deep road cutting. In thin section the rock was seen to be very fresh. No other pyroxenite site sampled gave a useful result, probably because of alteration with formation of secondary magnetite, visible in thin section. Directions of N.R.M. at this site have a significant mean on Watson's criterion, although k is only 1·7, because N is 26. Intensities of N.R.M. were low, and of the order of 10^{-5} c.g.s. units, compared with typical N.R.M. intensities $\sim 10^{-3}$ c.g.s. units at other sites. On demagnetization these cores did not reach a stable direction of magnetization. After washing in $H_{AC} = 150$ Oe peak, k increased to 4·8 and the mean direction remained near that of the N.R.M.

At Site 5 the directions of N.R.M. of seven cores were distributed along a great circle through the present geomagnetic field direction. The demagnetizing field of 300 Oe had to be chosen with some care (cf. Figure 2(b)). After the cores had been washed in this field, the directions of magnetization moved along the great circle to form a group. Two other cores showed anomalous directions of magnetization. Thin sections from these showed serpentinization and some secondary magnetite. A thin section from one of the seven cores first mentioned showed negligible alteration.

As was remarked in Section 3, Site 6 consisted of five widely spaced outcrops so chosen that it was improbable that lightning would have affected all the outcrops. Three cores from one outcrop showed scattered directions of N.R.M. which were unaffected by demagnetizing fields up to 800 Oe. At a second outcrop the directions of magnetization remained widely scattered after demagnetization. There is little doubt that lightning had affected these outcrops. Nine cores from the remaining three outcrops showed random directions of N.R.M. on Watson's criterion (Watson 1956). These cores showed a primary magnetization which was stable in direction for 600 Oe $\leqslant H_{AC} \leqslant$ 800 Oe. The high washing field of 600 gauss may indicate that these cores were partially affected by lightning I.R.M.

Two serpentinized picrite cores from Site 7 were rejected. Of the remaining cores from this site, five were picrite and three quartz gabbro. The five picrite cores and one quartz gabbro core gave closely grouped directions of N.R.M., and one of these showed little change for $H_{AC} \leqslant$ 800 Oe (Figure 2(d)). These six cores were washed in 300 Oe peak field. The other two quartz gabbro cores showed widely divergent directions of N.R.M., and required washing fields of 300 and 800 Oe respectively to bring them in to the group. The demagnetization of the latter core is shown in Figure 2(a). As has been remarked in Section 5, this core has probably been partially affected by lightning.

Sites 8 and 9 were in the two quartz gabbro satellite dykes and showed very similar magnetic properties. The N.R.M. directions were scattered at both sites. Demagnetization showed a primary component stable in direction for 200 Oe $< H_{AC} \leqslant$ 800 Oe (cf. Figure 2(c)). After washing in 300 and 400 Oe respectively, these sites gave exceptionally well grouped directions of primary magnetization (Figures 4(h), 3(b), Table 1).

At Sites 7, 8 and 9 only one specimen was measured from each core because of the excellent grouping of primary magnetizations, shown by the values of k in Table 1.

The site mean directions and radii of 95 per cent confidence are given in Table 1. Figure 5 shows the nine circles of confidence for the site mean directions of primary magnetization. It is clear that the nine sites, spread over 200 miles of the length of the Great Dyke and including five rock types, give highly consistent directions of primary magnetization. Clearly these directions are those of thermoremanent magnetizations acquired when the Great Dyke was intruded.

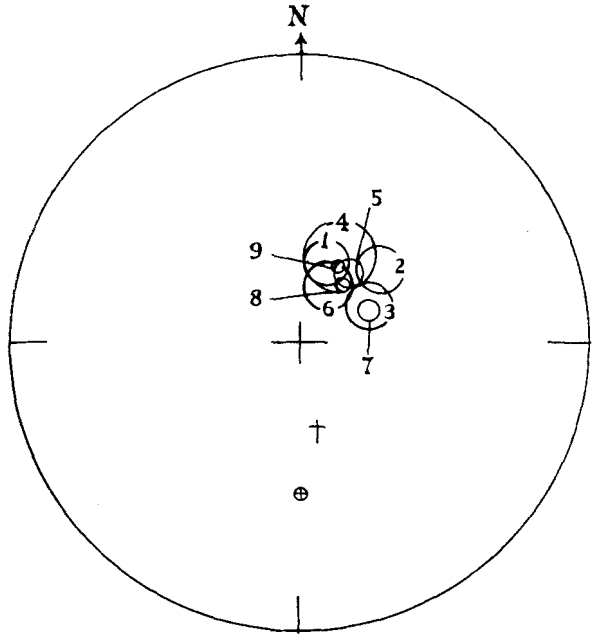


FIG. 5.—Circles of 95 per cent confidence for nine site mean directions of magnetization. Lower hemisphere stereographic projections, south seeking poles at all sites.

⊕—Axial dipole field (south-seeking)
†—Present field (south-seeking).

The nine site means taken with unit weight give a mean direction of magnetization $D = 218\frac{1}{2}°$, $I = -58\frac{1}{2}°$, $k = 68$, $\alpha_{95} = 6°$.

7. Displacement of rocks since magnetization

At those sites which are not near the centre of the Dyke the rocks have been tilted as a result of the general synclinal deformation, on Worst's theory (cf. Section 1). If the rocks cooled through their Curie points before the synclinal deformation developed, there should be a reduction in the dispersion of the site mean directions on applying appropriate corrections to restore the layers to the horizontal (Gough & van Niekerk 1959). For reasons given in Section 3, only at four of the sites used were the rocks appreciably tilted. Table 2 shows the strike and dip at each of these sites as measured from Worst's map (Worst 1960). The

Table 2

Attitudes of rocks at four sites

Site No.	Strike (E of N)	Dip
1	9°	15° ESE
2	9°	18° WNW
5	36°	12° SE
7	16°	25° ESE

parameters of the mean of the site mean directions with and without dip corrections, are given in Table 3; the symbols used have meanings as in Table 1. It will be seen that the mean directions are not significantly changed, but that the dispersion

Table 3

	D	I	k	α_{95}
Without dip correction	218½°	−58½°	68	6°
With dip correction	223½°	−57½°	35	8½°

is increased when dip corrections are applied. The reduction in k on applying dip corrections results, however, almost entirely from the large correction at a single site (Site 7). No definite answer can, therefore, be given to the question whether magnetization preceded or followed the synclinal deformation. The site mean directions uncorrected for dip have been used in calculating the pole position. This procedure is appropriate if the layers were originally horizontal and were deformed at a high temperature, as Worst suggests. It is, of course, equally appropriate if the rocks crystallized in their present attitudes.

8. Ancient African pole positions

On Worst's theory, the Dyke was intruded as a series of horizontal magma sheets of the order of 1000 ft in thickness. Such a sheet intruded at 1000 °C into country rock at 0 °C takes about 1800 years to cool to 600 °C at the centre, and is then at between 550 °C and 600 °C at the contacts (Jaeger 1957). Whatever the time interval between the injections of successive sheets, it is clear that the time interval covered in the cooling of a pile of five such sheets will not be less than 10^4 years. If the intrusions follow so rapidly that the Dyke approximates to a single intrusion at 1000 °C of thickness 4 miles, the centre will lag behind the margins 3×10^5 years in reaching 600 °C, or $2{\cdot}5 \times 10^5$ years in reaching 500 °C (Jaeger 1957). The sampling sites are distributed over 200 miles along the Dyke, over thousands of feet in depth, and between the centre and margins of the Dyke; and include three complexes and two satellite dykes. Whatever the detailed process of intrusion, it is highly probable that the sites became magnetized over a time interval of at least 10^4 years. It is probable, therefore, that the mean pole position approximates to the geocentric axial dipole position freed from secular variation.

It is possible to calculate the mean pole position from a number of site mean directions of magnetization, in one of two ways.

(1) A mean direction of magnetization is found by combining the site mean directions. Neglecting the variation in site positions, the pole position is found from the mean declination and inclination on the assumption of a geocentric dipole field. A cone of confidence is calculated for the mean direction of magnetization, on the assumption that the site mean directions are symmetrically distributed. When projected on to the surface of the Earth this results in an oval of confidence about the mean pole position. This oval is specified by the usual semi-axis parameters δm and δp (Irving 1956).

(2) A pole position is found corresponding to the mean direction of magnetization at each site. These pole positions may be combined to give a mean pole position with associated circle of confidence, using Fisher's statistics. The application of Fisher's statistics to the pole positions implies a symmetrical distribution of the poles, and a corresponding oval distribution of the directions of magnetization.

Creer, Irving & Nairn (1959) have applied the second method to thirty-four sites in the Whin Sill. The directions of magnetization were found empirically

to have an oval distribution. The choice between these two methods cannot, however, be made empirically where fewer sites are available.

In the absence of sufficient evidence concerning the manner in which the geomagnetic pole moves round the geographic pole, it is reasonable to assume that the secular variation produces, at instants distributed over 10^4 or 10^5 years, a symmetrical distribution of magnetic pole positions rather than an oval distribution. If the principal cause of between-site dispersion is secular variation, then the second method of computing the mean pole is to be preferred, especially where the sites are wide-spread as in the present case. If other causes such as orientation errors, movements of the rock or incomplete demagnetization predominate, then the first method is preferable. In the present work the orientation error is certainly small compared with secular variation, and at most sites it is clear that the effect of incomplete demagnetization on the site mean direction is small. It has already been suggested (cf. Section 1) that movements of the Dyke rocks involving rotation have been small at the sites used. It is probable that the secular variation accounts for most of the between-site dispersion. For convenience of comparison with other studies the mean pole has been calculated by both methods, and its parameters are given in Tables 4 and 5. The oval of 95 per cent confidence for the mean pole position as calculated by method (1), and the circle of confidence using method (2), are shown in Figure 6.

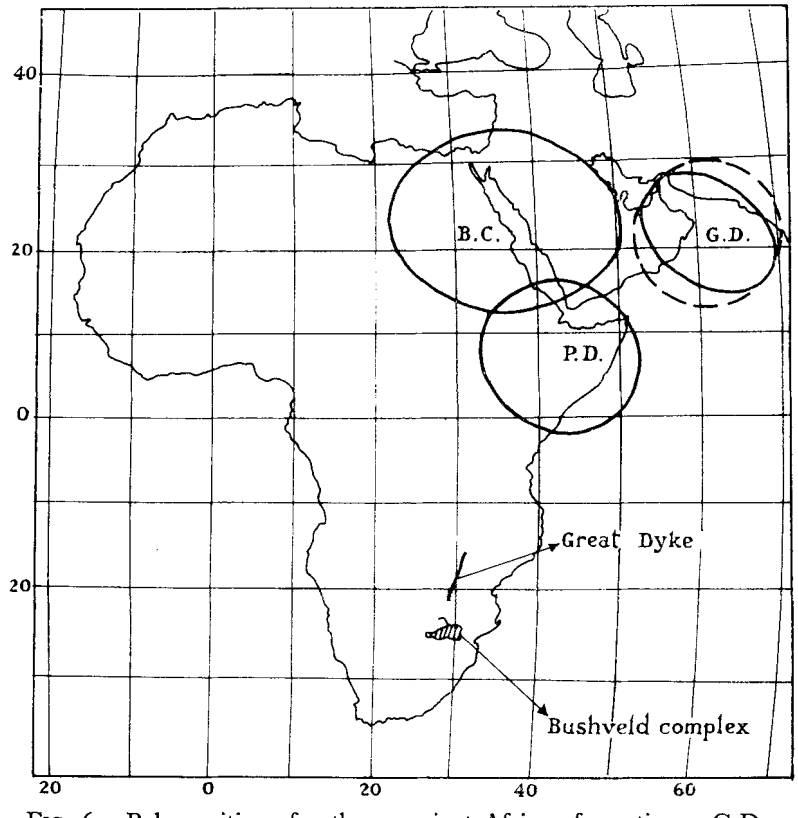


FIG. 6.—Pole positions for three ancient African formations. G.D.—Great Rhodesian Dyke, B.C.—Bushveld Complex, P.D.—Pilansberg Dykes.

Continuous lines show ovals of 95 per cent confidence. Broken line shows circle of 95 per cent confidence.

Table 4

Pole positions from three ancient African rock formations, calculated from mean directions of magnetization

Formation	N	D	I	R	k	α_{95}	Lat. N	Long. E	δm	δp	Pole
Great Dyke	9	218½°	−58½°	8·881	68	6°	21½°	61½°	9°	6½°	South
Bushveld Complex	5	10°	+59½°	4·941	68	9½°	23½°	36½°	14°	11°	North
Pilansberg Dykes	5	24°	+69½°	4·968	124	6°	7½°	42½°	10°	9°	North

N = No. of sites.
D, I, R, k and α_{95} are as defined in Table 1.
δp = radial semi-axis
δm = transverse semi-axis
} of oval of 95 per cent confidence for mean pole position.

Table 5

Pole positions from three ancient African rock formations, calculated by combining site pole positions

Formation	N	R	k	Lat. N	Long. E	α_{95}	Pole
Great Dyke	9	8·758	33	21½°	61½°	9°	South
Bushveld Complex	5	4·902	41	23°	36°	12°	North
Pilansberg Dykes	5	4·925	54	7½°	42½°	10½°	North

N = No. of site pole positions
R and k are as defined in Table 1.

α_{95} = Semi-axis of cone of 95 per cent confidence for mean pole position.

The pole positions given by the Bushveld gabbro (Gough & van Niekerk 1959) and by the Pilansberg Dykes (Gough 1956) are also given in Tables 4 and 5 and in Figure 6. The Bushveld Complex has been dated using rubidium–strontium and uranium–lead methods on total-rock and separated mineral samples from the granite, and uranium–lead methods on a sample from the gabbro (Schreiner 1958a, Nicolaysen, de Villiers, Burger & Strelow 1958). The results give concordant ages close to $2{\cdot}0 \times 10^9$ years. Schreiner (1958b) has dated biotite-bearing fractions of samples from one of the Pilansberg dykes at $(1{\cdot}29 \pm 0{\cdot}07) \times 10^9$ years. The Great Dyke has not yet been dated from radio-active decay. There are strong petrological similarities between the Bushveld Complex and the Great Dyke, and these have led various workers to suggest that they were contemporaneous (du Toit 1954, Worst 1960). Neither an upper nor a lower limit can at present be set to the age of the Great Dyke from stratigraphic relationships. It is hoped that rocks from the Great Dyke will be dated shortly. It will be seen in Figure 6 that apart from a geomagnetic reversal, the pole positions from the Pilansberg Dykes, the Bushveld Complex and the Great Rhodesian Dyke differ remarkably little. This is consistent with the view that the Great Dyke was intruded at about the same time as the Bushveld Complex.

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Note added in proof.—In a concurrent investigation of the palaeomagnetism of the Great Dyke, Dr A. E. M. Nairn (*Phil. Mag.*, in press) has obtained results in substantial agreement with those here reported.

Further note added in proof.—In recent work on samples of norite and quartz gabbro from the Great Dyke, using Rb^{87}–Sr^{87} decay, Faure has determined the age at $(2{\cdot}15 \pm 0{\cdot}30) \times 10^9$ years (*J. Geophys. Res.*, 1962, **67**, 3557). Faure assumed decay constant $1{\cdot}47 \times 10^{-11}$ yr^{-1}. With this decay constant those age determinations for the Bushveld Complex which depend upon Rb^{87}–Sr^{87} decay would require reduction by about 6 per cent.

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